

1. INTRODUCTION

1.1 Project Overview

Food choices and dietary habits play an important role in the health and well-being of college students. Understanding how students choose food and how their dietary habits relate to lifestyle and health factors can help identify important patterns and areas for improvement. This project, **“Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study,”** analyzes college students’ food choices using interactive Tableau visualizations. It examines breakfast habits, comfort food preferences, gender-based patterns, exercise, weight, healthy feeling, and other dietary and lifestyle factors. The insights generated through the analysis help understand student eating behavior and support the development of suitable dietary strategies and healthier lifestyle awareness.

1.2 Purpose

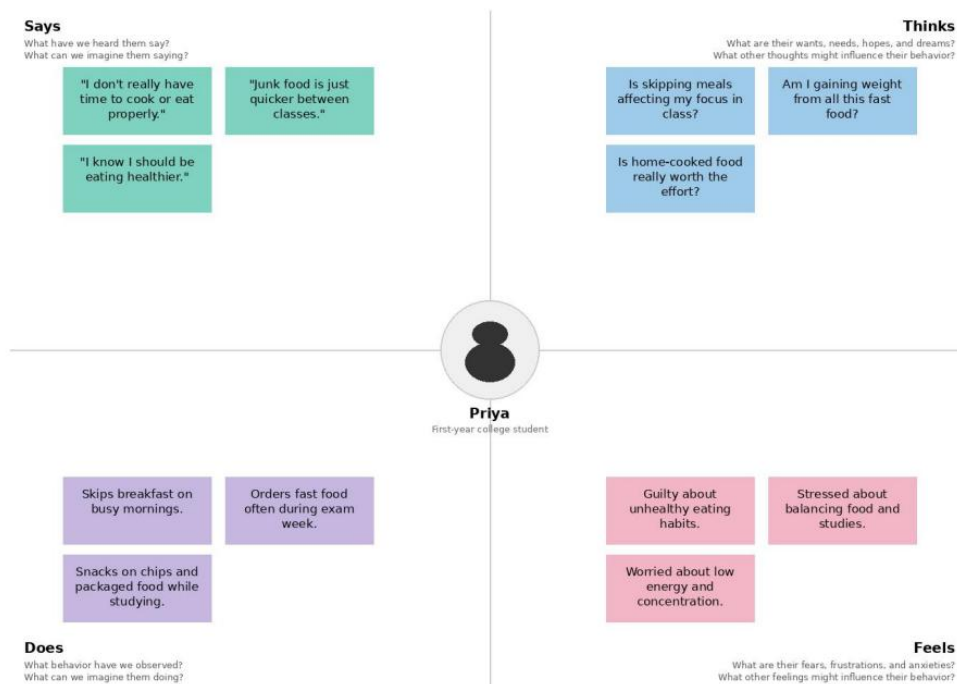
The purpose of this project is to analyze the food choices and lifestyle patterns of college students using Tableau. The project aims to identify patterns in dietary habits, examine relationships between food choices and health-related factors, and compare different student behaviors through interactive visualizations. It also seeks to provide meaningful insights through dashboards and a Tableau Story, helping students, educational institutions, and health-related stakeholders better understand dietary patterns and support healthier food and lifestyle decisions.

2. IDEATION PHASE

2.1 Problem Statement

I am	I'm trying to	But	Because	Which makes me feel
a second-year engineering student with a packed class schedule	stay focused and perform well academically	I often skip breakfast and rely on fast food between classes	I don't have time to plan or cook a balanced meal in the morning	sluggish, unfocused, and less productive during lectures
a first-year student living away from home during exam season	manage academic stress without compromising my health	I end up stress-eating comfort food and junk snacks late at night	I don't have healthy, quick alternatives easily available near my hostel	guilty, anxious, and low on energy the next day

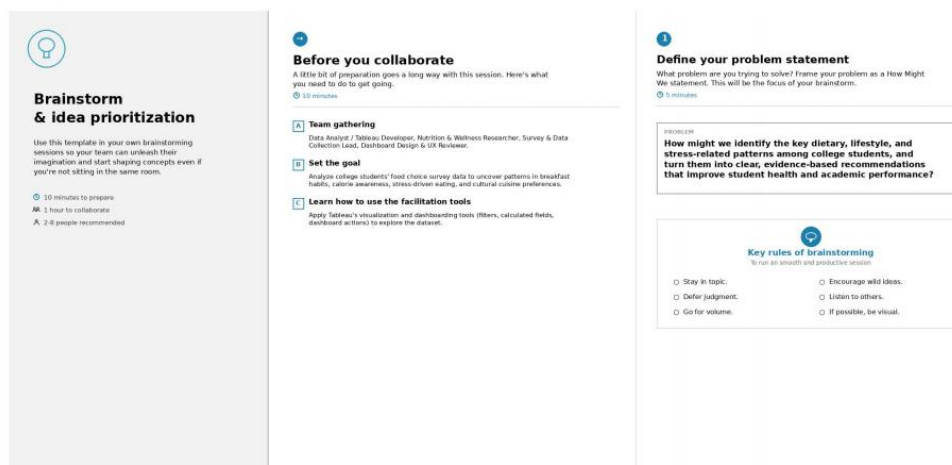
2.2 Empathy Map Canvas



2.3 Brainstorming

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Step-1: Team Gathering, Collaboration and Select the Problem Statement



Step-2: Brainstorm, Idea Listing and Grouping

2

Brainstorm

Write down any ideas that come to mind that address your problem statement.

10 minutes

Amar			Yuktesh			Person 3			Person 4		
Breakfast freq. vs. focus	Skipped meals in exams	Fast food vs. home-cooked	Comfort food reasons	Stress vs. snacking	Late-night eating	Fruit/veg intake by group	Calorie perceptions by area	Hydration habits	Cuisine prefs by region	Cafeteria vs. outside spend	Openness to healthy food

Person 5			Person 6			Person 7			Person 8		
Dashboard filter by year/level	Meal regularity vs. energy	KPI: balanced-diet %	Exercise frequency segments	Sleep habits vs. diet	Food-delivery app usage	Cafeteria recommendation dashboard	Healthier combo meals	Salad bar cost impact	Stress-eating triggers etc.	Wellness workshop topics	Student awareness infographic

3

Group ideas

Take turns sharing your ideas while clustering similar or related notes as you go. Give each cluster a sentence-like label.

20 minutes

1. Meal Regularity & Academic Focus

- Breakfast frequency vs. concentration/focus
- Skipped meals during exams
- Fast food vs. home-cooked comparison

2. Stress & Comfort Eating

- Reasons for comfort food consumption
- Stress vs. snacking correlation
- Late-night eating patterns

3. Nutrition & Calorie Awareness

- Fruit and vegetable intake tracking
- Calorie awareness by year of study
- Hydration and beverage choices

4. Cultural Preferences & Spending

- Cuisine preference by region
- Cafeteria vs. outside food spend
- Openness to healthier alternatives

5. Cafeteria & Wellness Recommendations

- Healthier combo-meal suggestions
- Salad bar cost-impact modeling
- Wellness workshop topics from data

Step-3: Idea Prioritization

4

Prioritize

Your team should all be on the same page about what's important moving forward. Place your ideas on this grid to determine which ideas are important and which are feasible.

20 minutes

Importance	High	Medium	Low
High	Real-time stress-eating trigger dashboard		Breakfast habits vs. academic focus dashboard
Medium	Institution-wide behavior change program		Stress vs. comfort food analysis
Low	Cultural cuisine preference mapping		Cafeteria healthy-options recommendation
	Exercise & sleep vs. diet segmentation		Calorie & nutrition awareness KPI cards
			Pro/vegetable intake visualization

Feasibility

Regardless of their importance, which tasks are more feasible than others? (cost, time, effort, complexity, etc.)

✓

After you collaborate

You can export the mural as an image or PDF to share with members of your team or instructors who might find it helpful.

Quick add-ons

Share the mural

Make a live link to the mural with the outcomes of the session.

Export the mural

Export a copy of the mural as a PNG or PDF to attach to emails, tickets, or your submission.

Keep moving forward

Strategy blueprint

Define the components of a new idea or strategy for the next project phase.

Customer experience journey map

Understand student needs, motivations, and pathways for a better experience.

Strengths, weaknesses, opportunities & threats

Identify strengths, weaknesses, opportunities and threats (SWOT) to develop a plan.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

PROJECT JOURNEY ▶

Customer Journey Map

Healthy Lifestyle and Dietary Habits Analysis using Tableau



3.2 Solution Requirement

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Collection & Preparation	Collect the College Food Choices dataset, connect the dataset to Tableau, validate records, and prepare data for analysis.
FR-2	Data Preparation	Handle missing values, clean and format data, verify data types, and prepare fields for visualization.
FR-3	Data Visualization	Create charts to analyze gender, breakfast habits, food choices, comfort food reasons, exercise, vitamins, and healthy feeling.
FR-4	Dashboard	Create an interactive dashboard, display key insights, use Gender, Breakfast, and Healthy Feeling filters, and support interactive exploration.
FR-5	Story	Create a 5-scene Tableau Story, present major insights in sequence, and highlight dietary and lifestyle patterns.
FR-6	Performance Testing	Verify data rendered, utilization of filters, calculated fields, and the number of visualizations/graphs.
FR-7	Web Integration	Publish the Tableau dashboard and embed the dashboard and Story using Flask with a responsive interface.
FR-8	Project Demonstration & Documentation	Record the project explanation video and prepare step-by-step project documentation.

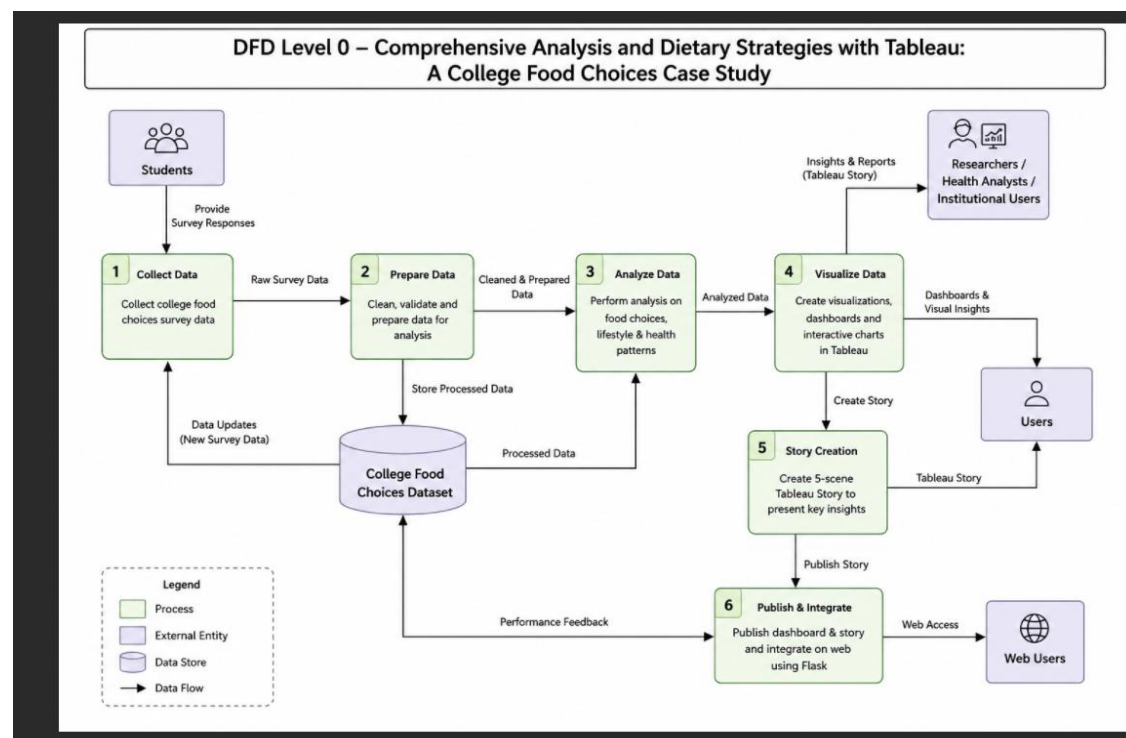
Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

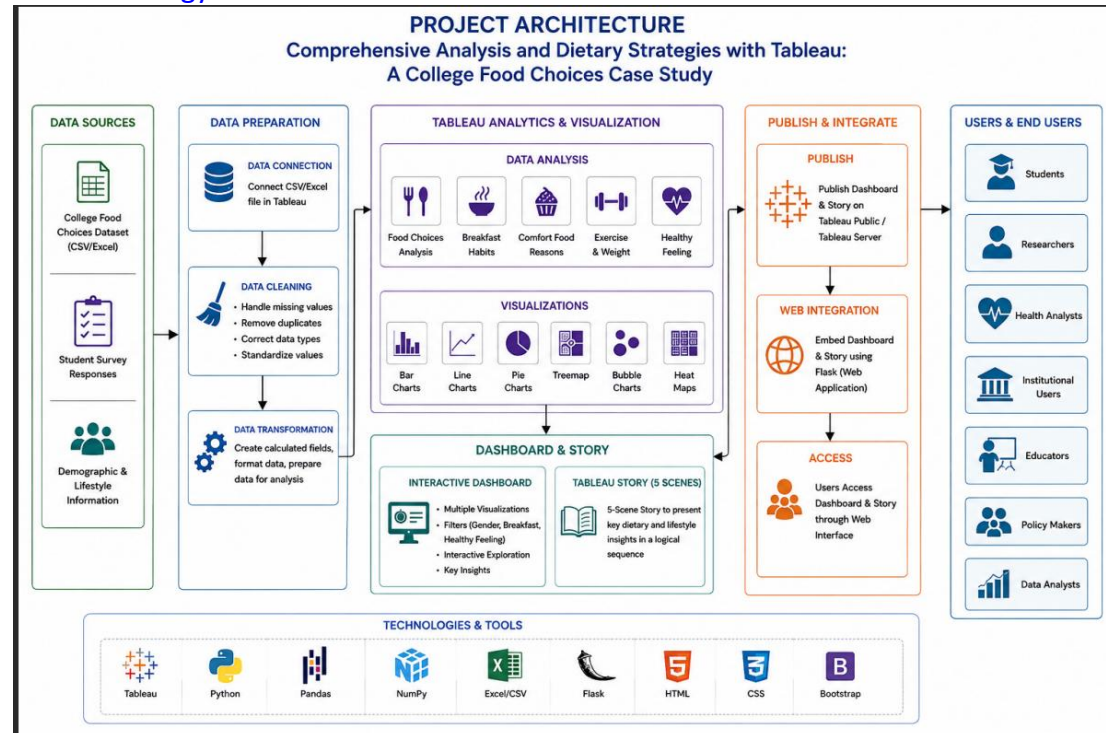
FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The Tableau dashboard and visualizations should be easy to understand and interact with.
NFR-2	Security	The analysis should accurately represent the College Food Choices dataset.

NFR-3	Reliability	The dashboard, filters, and visualizations should function consistently and provide reliable insights.
NFR-4	Performance	The Tableau dashboard and worksheets should load and respond efficiently.
NFR-5	Availability	The published dashboard should be accessible to intended users when deployed.
NFR-6	Scalability	The solution should support larger datasets and additional dietary or lifestyle variables in future.

3.3 Data Flow Diagram



3.4 Technology Stack



4. PROJECT DESIGN

4.1 Problem Solution Fit

Problem-Solution Fit Canvas			Purpose / Vision:	To analyze college students' food choices and lifestyle patterns and provide dietary strategies through interactive Tableau visualizations.	Version:	1.0
Define CS, RI, into CL	1. CUSTOMER SEGMENT(S) CS	6. CUSTOMER LIMITATIONS CL EG. BUDGET, DEVICES	5. AVAILABLE SOLUTIONS AS PROS & CONS			
	- College students - Students interested in healthy eating - College health and wellness teams - Nutrition and dietary researchers - Educational institutions	- Limited nutritional knowledge - Lack of time for detailed analysis - Difficulty interpreting raw data - Limited awareness of healthy habits - Data may not represent all students	- General nutrition websites - Diet and calorie tracking applications - Health and fitness applications - Manual survey analysis - Traditional statistical reports			
Focus on PR, top into BE, understand RC	2. PROBLEMS / PAINS PR + ITS FREQUENCY	9. PROBLEM ROOT / CAUSE RC	7. BEHAVIOR BE + ITS INTENSITY			
	- Irregular breakfast and meal habits - High fast-food consumption - Unhealthy comfort-food choices - Stress and boredom-related eating - Lack of awareness of healthy dietary patterns - Difficulty understanding lifestyle and health relationships	- Lack of awareness about dietary habits - Busy college lifestyle and schedules - Stress-related eating behavior - Convenience of fast food - Limited time for meal preparation - Different lifestyle and exercise patterns	- Skip or vary breakfast habits - Consume fast food - Eat comfort food during stress or boredom - Follow different exercise routines - Search online for diet and health information - Use health and fitness applications			
Identify Actions (TR & EM)	3. TRIGGERS TO ACT TR	10. YOUR SOLUTION SL	8. CHANNELS OF BEHAVIOR CH			
	4. EMOTIONS EM BEFORE / AFTER Before: - Unaware of unhealthy food habits - Concerned about weight and health - Confused about dietary choices After: - Better awareness of food habits - Understanding of important patterns - Support for healthier choices	- Analyze college food choices and lifestyle data - Prepare and validate data using Tableau - Create interactive visualizations - Analyze breakfast, comfort food, exercise, weight and health patterns - Develop an interactive dashboard - Use filters for better exploration - Present insights through a 5-scene Tableau Story - Support dietary awareness and healthier decision-making	- College health programs - Social media - Nutrition and health websites - Fitness applications - Educational platforms - Online search engines			

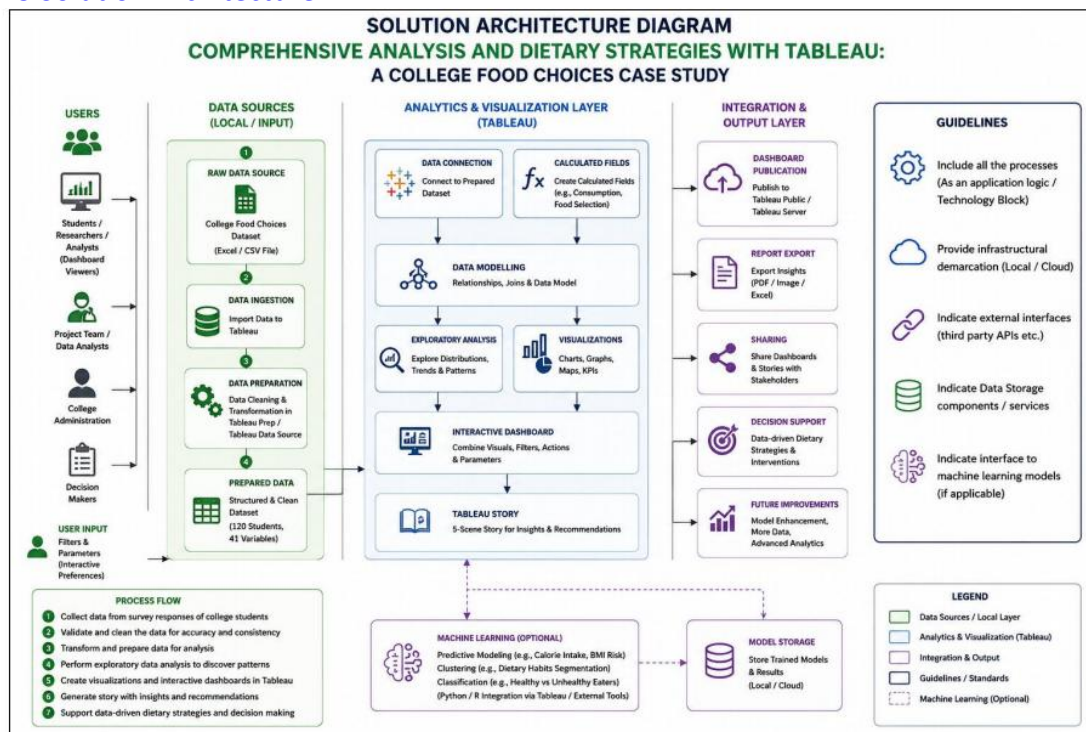
4.2 Proposed Solution

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	College students have varied food and lifestyle habits. The project analyzes dietary patterns to identify factors affecting healthy food choices.
2.	Idea / Solution description	Develop an interactive Tableau dashboard to analyze food choices, breakfast, comfort food, exercise, weight, and health patterns.
3.	Novelty / Uniqueness	Combines multiple dietary and lifestyle analyses in one Tableau dashboard with interactive filters and a 5-scene Story.
4.	Social Impact / Customer Satisfaction	Helps students and institutions understand dietary habits and supports healthier food choices and wellness awareness.
5.	Business Model (Revenue Model)	Can be offered as a customized analytics and reporting solution for educational and health organizations.
6.	Scalability of the Solution	Can be expanded with larger datasets, multiple colleges, real-time health data, and predictive analytics.

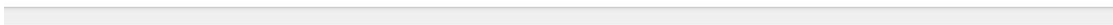
4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection & Preparation	USN-1	As a team member, I can collect and connect the College Food Choices dataset to Tableau.	3	High	S. Sadaquunnisa B.Nikitha
Sprint-1	Data Preparation	USN-2	As a team member, I can clean and prepare the dataset for analysis and visualization.	3	High	B.Nikitha
Sprint-1	Data Visualization	USN-3	As a user, I can view visualizations of college food and lifestyle patterns.	4	High	B. Nikitha,
Sprint-2	Dietary Analysis	USN-4	As a user, I can analyze breakfast, food choices and comfort-food patterns.	4	High	S. Sadaquunnisa
Sprint-2	Dashboard	USN-5	As a user, I can explore an interactive Tableau dashboard using filters.	6	High	S. Sadaquunnisa
Sprint-1	Health & Lifestyle Analysis	USN-6	As a user, I can analyze exercise, weight and healthy-feeling patterns.	4	Medium	B.Nikitha
Sprint-1	Tableau Story	USN-7	As a user, I can view project insights through a structured 5-scene Tableau Story.	6	High	B.Nikitha



Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-2	Performance Testing	USN-8	As a team member, I can verify data loading, filters, calculations and visualizations.	4	High	S. Sadaquunnisa
Sprint-2	Web Integration	USN-9	As a user, I can access the published Tableau dashboard through web integration.	3	Medium	S. Sadaquunnisa
Sprint-2	Documentation	USN-10	As a team member, I can prepare project documentation and project demonstration	3	High	T. Sadaquunnisa B.Nikitha

5.2

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	10	3 Days	27 Jun 2026	29 Jun 2026	20	29 Jun 2026
Sprint-2	20	2 Days	30 Jun 2026	01 Jun 2026	10	01 Jun 2026
Sprint-3	10	3 Days	02 Jul 2026	02 Jul 2026	10	04 Jul 2026
Sprint-4	20	3 Days	07 Jul 2026	07 Jul 2026	10	07 Jul 2026

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

Total Sprint Duration: 11 Days

Average Velocity:

Velocity = Total Story Points / Total Duration

Velocity = 40 / 11 = 3.64 ≈ 4 Story Points per Day

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

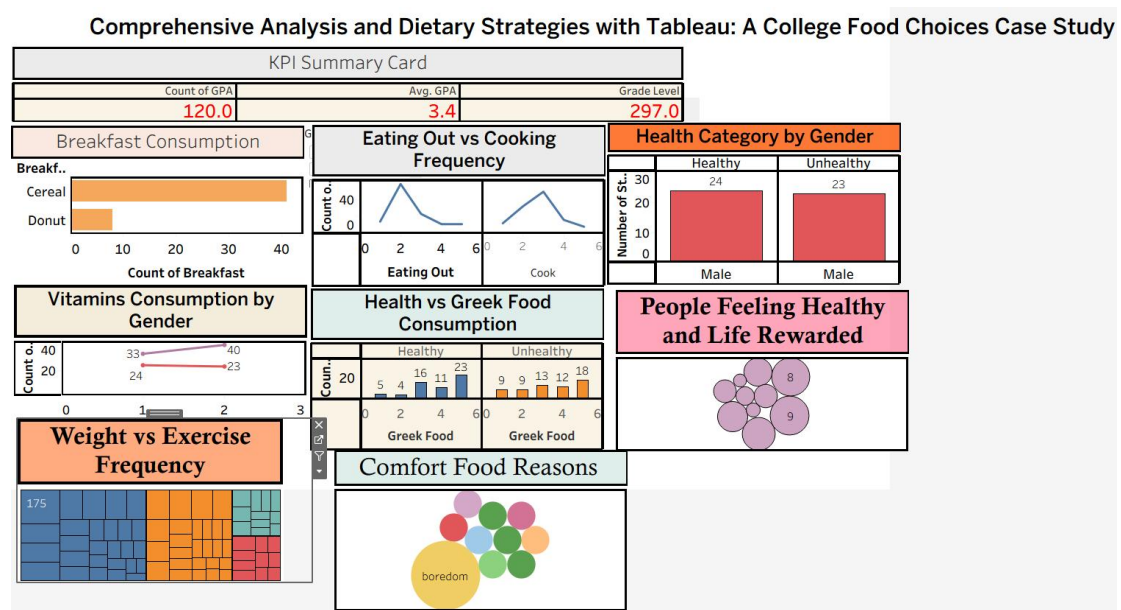
6.2

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	College Food Choices dataset successfully loaded into Tableau. The dataset contains 120 student records and 41 variables , including dietary, lifestyle, demographic, health, and academic-related information.
2.	Data Preprocessing	Removed null values where required, verified data types, cleaned and formatted fields, and prepared the dataset for visualization and analysis.
3.	Utilization of Filters	Filters used: Gender, Breakfast, and Healthy Feeling.
4.	Calculation fields Used	Used calculated fields required for analysis and data visualization. Calculations supported counts, comparisons, and analytical insights in the Tableau worksheets.
5.	Dashboard design	Interactive dashboard created using multiple visualizations to analyze food choices, gender, breakfast habits, comfort food reasons, exercise, vitamins, and healthy feeling. Filters provide interactive exploration.
6.	Story Design	7 Tableau Story scenes created to present the major analyses and insights in a logical sequence, covering student dietary habits, lifestyle patterns, health-related factors, and findings.

6.3

7. RESULTS

7.1 Output Screenshots



Dashboard :

<https://public.tableau.com/app/profile/nikitha.banda/viz/Foodchoices42/CollegeFoodChoicesDashboard?publish>

Story:

<https://public.tableau.com/app/profile/nikitha.banda/viz/Foodchoices42/CollegeFoodChoicesStory?publish>

1. ADVANTAGES & DISADVANTAGES

Advantages

- Provides an interactive way to analyze college students' food and lifestyle patterns.
- Combines multiple analyses such as breakfast habits, food choices, comfort food, exercise, weight, and healthy feeling.
- Interactive filters help users explore data based on different categories.
- The Tableau dashboard presents complex data in an easy-to-understand visual format.
- The 5-scene Tableau Story presents important insights in a logical sequence.
- Helps students and institutions identify dietary patterns and promote healthy lifestyle awareness.
- The solution can be extended with larger datasets and additional variables.

Disadvantages

- The analysis is limited to the available College Food Choices dataset.
- A relatively small dataset may not represent all college students.
- Results depend on the accuracy and completeness of the collected data.
- The project provides analytical insights but does not provide real-time health or dietary monitoring.
- Some visualizations may require users to have basic knowledge of dashboards and filters.
- The project does not currently include predictive analysis or personalized recommendations.

Conclusion

The **Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study** successfully analyzes college students' dietary and lifestyle patterns using interactive Tableau visualizations. The project connects and prepares the dataset, analyzes important factors, and presents insights through visualizations, an interactive dashboard, and a 5-scene Tableau Story. The analysis helps in understanding food choices, breakfast habits, exercise, weight, comfort food preferences, and healthy-feeling patterns. Overall, the project demonstrates how data visualization can support better understanding of student dietary behavior and promote healthier food and lifestyle awareness.

Future Scope

- Integrate data from multiple colleges and a larger number of students.
- Include real-time data collection through web or mobile applications.
- Add more dietary, fitness, and health-related variables.
- Develop predictive models to identify future dietary and lifestyle trends.
- Provide personalized food and wellness recommendations.
- Integrate nutritional information and food databases.
- Improve the web interface and provide mobile access.
- Add advanced analytics to support institutions, researchers, and health organizations.

11. APPENDIX

Source Code (if any)

Dataset Link:

https://drive.google.com/file/d/1AmQoxCQaxi0wujkfaZ8jSVxSoJbGDvoA/view?usp=drive_link

GitHub & Project Demo Link:

<https://github.com/Shaiksadaqunnisa/Comprehensive-Analysis-Dietary-Strategies-with-Tableau-A-college-food-choices-case-study>

<https://drive.google.com/file/d/1ZXBvMUcj0d9F2Jv7rMVsq8SOmcBkfKuc/view>